



Model-level attention and batch-instance style normalization for federated learning on medical image segmentation

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ABSTRACT

Federated learning (FL) offers an effective privacy protection mechanism for cross-center medical collaboration and data sharing. In multi-site medical image segmentation, FL allows each medical site to act as a client, forming its own data domain. FL has the potential to enhance the performance of models on known domains. However, practical deployment faces the challenge of domain generalization (DG) due to the non-identical and non-independent (non-IID) nature of data from different domains. This results in decreased model performance in unseen domains. Current DG solutions are overly complex in addressing style differences and lack focus on inter-domain image features causing model differences. Furthermore, these solutions are not suitable for the FL paradigm that requires data storage separation. Hence, the lightweight model-level attention and batch-instance style normalization (MLA-BIN) is proposed to solve the DG of FL in this study. The MLA module represents the unseen domain as a linear combination of seen domain models. It does not require access to raw data but learns from the sufficient exploration of data features in known domains, thereby identifying differences in inter-domain data features and enabling the global model to generalize from seen to unseen domains. In the BIN block, batch normalization (BN) and instance normalization (IN) are combined to perform the shallow layers of the segmentation network for style normalization. By integrating the segmentation backbone network with the BIN block (BIN-Net), it ensures effective learning of intra-domain features and addresses the impact of inter-domain image style differences on domain generalization without accessing data from other centers. Extensive experimental results demonstrate that the proposed method achieved a Dice similarity coefficient of 88.27, 88.25 and 64.94 on the prostate, the optic disc and cup, and the COVID-19 lesion segmentation dataset, respectively, outperforming the state-of-the-art methods.

Abbreviations

Federated learning FL
Domain generalization DG
Model-level attention MLA
Batch-instance style normalization BIN
Batch normalization BN
Instance normalization IN
Convolutional neural networks CNN

1. Introduction

In deep learning for medical images, there is a desire to use multi-center data to build more accurate prediction and diagnosis models while ensuring the protection of user privacy in cross-center medical collaboration [1,2]. Federated learning (FL) [3] provides a balanced solution between these two requirements. FL consists of a server and multiple clients at different centers, as shown in Fig. 1. During the training process, the server first distributes the global model to each client, and then each client uses its own center's data to train a local model. Afterwards, the locally trained models are uploaded to the

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server. Finally, the server aggregates these models and obtains a new global model to complete one round of training. In FL, each client independently collects, annotates, and uses local data, and datasets from different centers are defined as different domains. Therefore, a global model with cross-domain features and higher accuracy can be obtained through multiple rounds of iterative training.

Despite the effectiveness of FL in utilizing multi-center data [4,5], current federated models underperform in the generalization of unseen domain data known as the domain generalization (DG) problem. This limitation is particularly evident in clinical deployment, where variations in imaging instruments and protocols in different medical centers lead to disparities in image data and features of convolutional neural networks (CNN) trained on data from different centers. These problems lead to heterogeneity in medical data. These challenges make it difficult for the model to achieve good application performance in newly deployed institutions. The need for data annotation arises when the model cannot produce the desired segmentation in a new institution, a practice that is both time-consuming and impractical [6]. The potential of FL in practical applications is limited by the low reusability of such models and their inability to be used for in-depth clinical research. Hence, solving the DG problem in FL is the key to realizing the practical applications of FL. It is also the fundamental purpose of promoting medical data collaboration and sharing.

Recently, novel DG solutions have emerged under the FL paradigm [6–9]. For feature differences, the current solutions involve utilizing generative adversarial networks (GAN) to align features from different domains [6] and applying contrastive learning to correct the local model

feature deviations [8]. For style differences, the primary solutions comprise of enforcing style feature invariance [7], data augmentation [10], and meta-learning paradigms [9]. However, these solutions solely focus on problem-solving, neglecting the practical deployment issues. Specifically, meta-learning and GAN are likely to impose a heavier client burden. This is especially true for real-world situations such as in medical institutions, where the GPU configurations are limited, thus necessitating lightweight solutions.

To address the DG problem in FL, we propose a novel lightweight approach that adapts the deep and shallow layer features of images to the unseen domain. This approach aims to eliminate features and style differences presented between seen and unseen domains. Specifically, a linear combination of a set of seen domain models is used to represent the global model of an unseen domain. During the testing phase in the unseen domain, the weights of each seen domain model are adaptively adjusted based on the similarity between the deep layers features of the unseen and seen domains. At the same time, features in the shallow layers of the model are adjusted according to the network architecture to achieve style normalization of different domain images, as shown in Fig. 1.

The motivation of our approach stems from the observation that FL learns cross-domain features by combining models from each client. However, using fixed weight parameters based on data quantity will lead to the global model being affected by the quantity of data in each client. For example, if the domain with a large number of images does not match the feature of images in the unseen domain, it will lead to the decrease in the performance of the global model in the unseen domain.

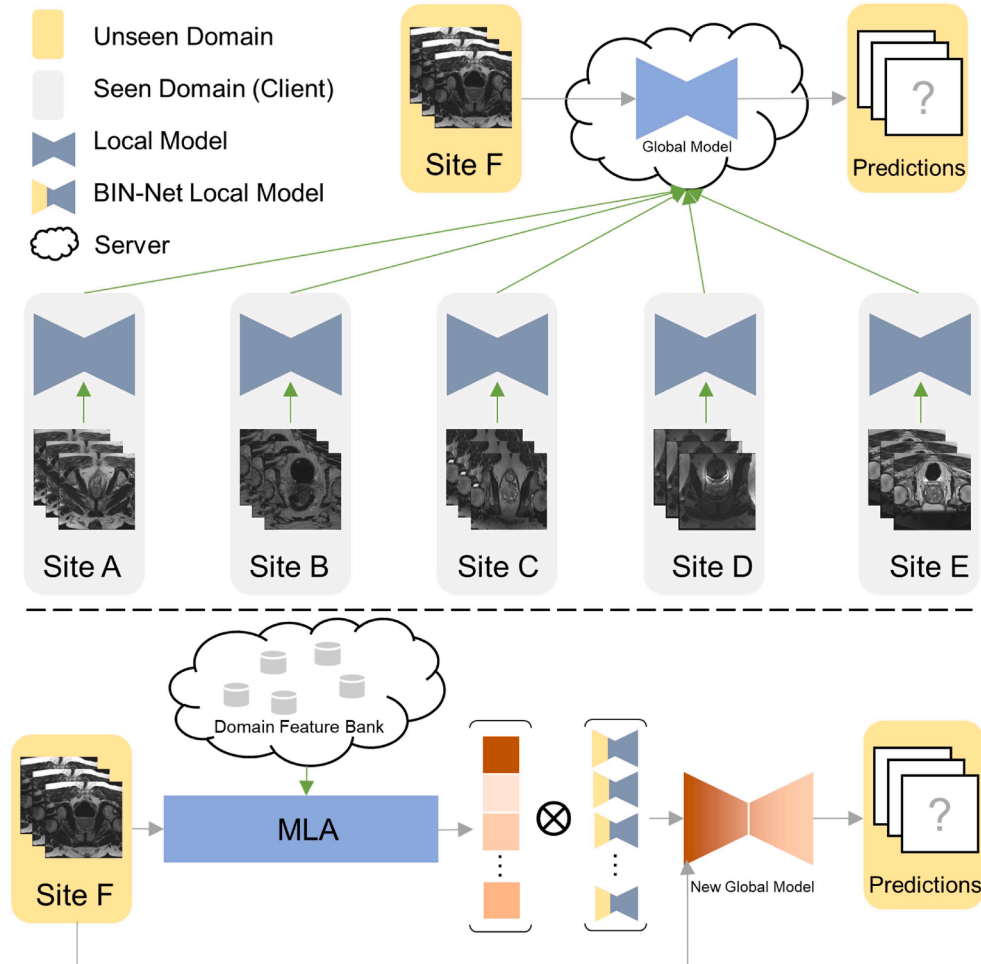


Fig. 1. Illustration of traditional FL training method and FL model testing on unseen center (top) and our proposed MLA-BIN model (bottom). The green arrow denotes the training phase, while the gray arrow denotes the testing phase.

Therefore, an attention mechanism that adaptively assigns weight parameters to each client based on the similarity of features between the unseen and seen domains is designed. Following the findings of style transfer studies [11,12], it has been observed that the CNN architecture stores style information in the statistical characteristics of each feature channel in the shallow layers of the network (i.e., mean and variance). Instance normalization (IN) can normalize the statistical information of each feature channel, while batch normalization (BN) can prevent internal covariate shift during the training process. Therefore, a block using both BN and IN in the shallow layers of the network is devised to improve the performance of the local model and eliminate the style differences between domains.

Our main contributions are as follows:

- We proposed a model-level attention module (MLA) that is suitable for FL. The MLA represents the unseen domain as a linear combination of seen domain models and adaptively adjusts the weights of seen domain models through an attention mechanism, thereby enabling generalization to the unseen domain.
- We designed a batch-instance style normalization block (BIN) that is suitable for FL to solve the domain gap caused by style differences. It is combined with the segmentation backbone network (BIN-Net), which can not only effectively learn intra-domain features but also eliminate inter-domain style differences without requiring access to data from other centers.
- We conducted extensive experiments and presented experimental results in three medical image segmentation tasks. Our proposed method outperformed state-of-the-art DG methods in FL. The effectiveness of our method was demonstrated through in-depth analytical experiments.

2. Related work

2.1. Federated learning in medical imaging

FL [3,13] is a collaborative machine learning method whereby multiple clients, such as medical institutions, share a model while preserving data privacy. The technique holds a wide potential for applications in medical imaging, enabling analysis of medical data across institutions, regions, and modalities to support medical diagnosis, prediction, and treatment [14–19]. Sheller et al. [18] conducted a pilot study to examine a collaborative model training method for multi-site brain tumor segmentation without data interaction. Jiménez-Sánchez et al. [20] proposed an FL method based on memory-aware curriculum learning for a breast cancer classification task involving multiple centers with exciting results. Wang et al. [19] introduced an FL and blockchain-based COVID-19 auxiliary diagnosis method. These studies leverage data resources from multiple medical institutions, effectively addressing the problem of data silos and improving model accuracy. However, these studies focus primarily on the value of FL within multiple sites, neglecting the problem of weak model generalization ability (i.e., DG) in actual clinical applications when facing unseen sites. In contrast, this paper focuses on improving the generalization ability of FL model in unseen domain.

2.2. Domain generalization

Numerous studies have proposed effective solutions to enhance the generalization ability of models for resolving DG problems [21–24]. Currently, the methods proposed in DG [25–30] are mainly based on centralized datasets. One of the most comprehensible techniques is to apply GAN for learning the stylistic features of images in the source domain and transferring learned style to the target domain [31,32]. Nevertheless, this approach could cause content information loss and interference in practical applications. Another conventional approach focuses on improving model robustness via extensive data augmentation

[10]. However, the practicality of this approach in real-life contexts is limited because it requires experience settings to upgrade the generalization ability in a fresh environment. Novel adaptive solutions have attracted considerable research [22,23,33,34]. Nonetheless, the use of data from multiple centers is required during the training phase of these methods. Given that safeguarding data privacy is crucial for FL (i.e., data is distributed across different centers), such methods are infeasible and unsuitable for the FL paradigm.

Pan et al. [25] proposed the use of IN for inter-domain image style normalization by studying the role of IN in CNN. However, their method was not successful in achieving complete style normalization via concatenation of IN and BN. Attention mechanisms have recently been explored in DG due to their ability to enhance network focus on target regions and feature expression. Gu et al. [26] proposed a DG solution based on domain composition, which effectively solved the DG problem via an attention mechanism that assigns weights to channels within the CNN. Hu et al. [27] proposed a solution using dynamic convolutions of domain and content. However, these methods require data from different domains to be centralized on the server, which is not consistent with the FL paradigm. Recently, solutions for DG within the FL paradigm have been developed [35–39]. Liu et al. [36] explored a solution based on FL using the principle of amplitude spectrum and meta-learning. Zhang et al. [37] solved the DG problem by aligning the feature distribution from different domains using a federated GAN. Nonetheless, the use of such techniques may escalate the costs of training in the FL setting. In contrast, our paper proposes a lightweight solution in which MLA is utilized to ensure even data storage distribution.

3. Method

3.1. Problem definition and method overview

Let $D = \{D_1, D_2, \dots, D_k\}$ be a set of source domains, with an unseen domain denoted as D_u . Each domain represents a data center and serves as a client in the FL setting. The t th image and label in the D_n domain are denoted as x_n^t and y_n^t , respectively.

To enable FL to generalize to an unseen domain, the MLA is proposed that utilizes image features to detect affiliated domain, as shown in Fig. 2(a). By considering the similarity of image features between seen and unseen domains and learning cross-domain features, the MLA avoids negative transfer effect and achieves a global model with high segmentation performance in the unseen domain, as shown in Fig. 2(b). Inspired by style transfer, the BIN block for the shallow layers of segmentation network is designed to enable style normalization of different center data, as shown in Fig. 2(c).

3.2. Model-Level attention

According to the definition of the aggregation phase in FL, the aggregation formula for the global model can be formally expressed as

$$G(m) = \sum_{i=0}^k w_i l_i(m) \quad (1)$$

where $l_i(m)$ represents the local model of the D_i domain, w_i represents the weight of the D_i domain, and $G(m)$ represents the global model obtained by weighted summation of k clients.

Therefore, the aggregation can be expressed as a linear combination of models from seen domains to develop a global model using test data from unseen domain. The traditional aggregation methods use fixed values for weighting coefficients $w = \{w_1, w_2, \dots, w_k\}$, such as the ratio of the center data quantity to the total data quantity. Nevertheless, in the case of test data from unseen domains, the difference in data features among distinct domains affects the weighting coefficient and results in sub-optimal global models. Therefore, an attention mechanism is introduced to obtain optimal weights of models based on the similarity

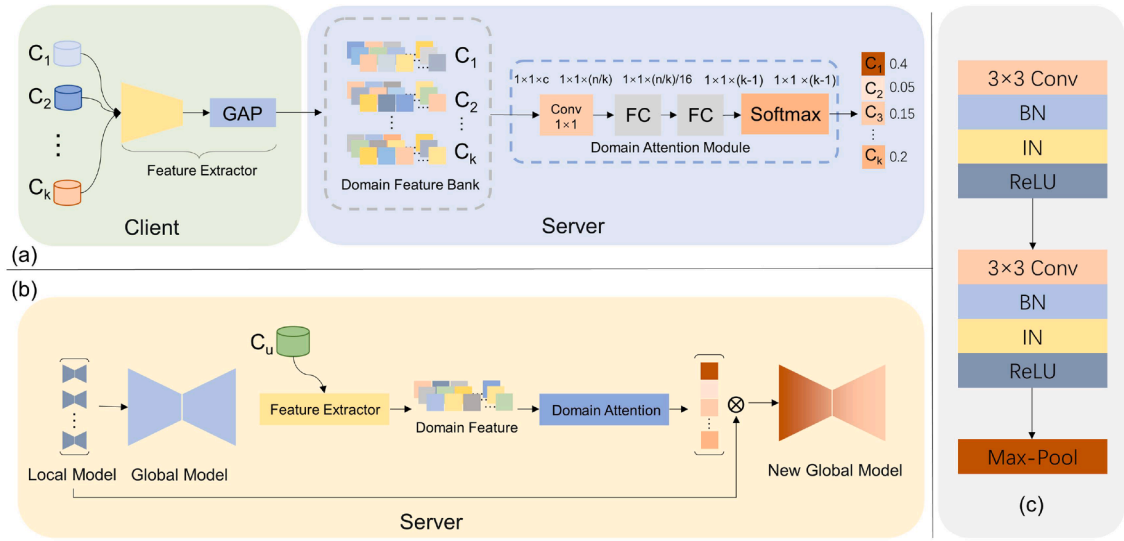


Fig. 2. Overview of the proposed model-level attention (MLA) and batch-instance style normalization (BIN) block. (a) Details of the proposed MLA training phase. (b) Details of the proposed MLA application phase. (c) Structure of the BIN block.

of inter-domain data features and enable the global model to generalize to the unseen domain.

In addition, traditional attention methods are designed for network-level attention by assigning weights to each feature channel in the network architecture. In contrast, MLA is more suitable for FL since it weights models of each domain instead of feature channels within the network architecture.

The MLA consists of three parts: feature extractor, domain feature bank, and domain attention module (DAM). The feature extractor is utilized to extract deep layer features from image data using CNN on the clients. The domain feature bank is used to store the image features on the server-side. The Domain Attention Module (DAM) is employed to identify the affiliated domain of the image features. The operational process of MLA will be elaborated below.

3.2.1. Training

MLA implements output at the model level, thus requiring a separately trained attention module as shown in Fig. 2(a). The purpose of MLA is to identify similarities between input data and data from seen domains. Sharing seen domain images is prohibited in FL. Hence, it is proposed to use CNN as a feature extractor to extract deep features from images. Specifically, a pre-trained CNN model with unified pre-training weights is first used to extract features from data in each client $D = \{D_1, D_2, \dots, D_k\}$. In this study, we utilized the ResNet-101 backbone architecture [40] as a pre-trained model for feature extraction. The pre-trained model is used to perform feature extraction on each image. The extracted features represent the deep representations learned by the pre-trained model from the images. Different images will have distinct deep representations due to variations in their features. Next, a global average pooling operation is performed on the extracted features, resulting in a mapping of images to the feature space. Finally, the extracted features from each client are uploaded to the server to construct a shared domain feature bank. The formula for mapping images to the feature space can be expressed as

$$C_n^t = \text{GAP}(\text{Enc}(x_n^t; \theta_{\text{Enc}})) \quad (2)$$

where GAP represents global average pooling layer, Enc represents the encoder part of the CNN with pre-trained model, θ_{Enc} represents the encoder parameters, and C_n^t represents the representation of the t th image in the D_n domain mapped to the feature space. The domain feature bank data is employed as the dataset to train MLA. These data are used as input features and the corresponding labels to train a

domain-multilayer perceptron as DAM on the server. The features are optimized to be maximally representative of each domain through a process of DAM. The DAM is designed to identify and emphasize the unique characteristics of each domain. The DAM consists of a 1×1 convolutional layer and two fully connected layers. It outputs k values through the soft-max function, denoted as D^w , where k values correspond to the probabilities that the input features belong to k domains. D^w satisfies $\sum_{i=1}^k D_i^w = 1$. The details of each layer are shown in Table 1. The convolutional layer and fully connected layers interact with the extracted feature maps from the base model, optimizing for the domain-specific objectives. This process enhances the encoder's ability within the DAM to strengthen the representation capacity of features for their respective domains. The formula for the DAM can be expressed as

$$D^w = \text{SM}(\text{FC}_2(\text{FC}_1(\text{Conv}(C_n^t; \theta_C); \theta_{\text{FC}_1}); \theta_{\text{FC}_2})) \quad (3)$$

where SM , FC and Conv respectively represent the soft-max layer, fully connected layer, and convolutional layer. θ_C and θ_{FC} respectively represent the trainable parameters of the convolutional and fully connected layers.

3.2.2. Application

To test the model in an unseen domain on the server, the trained DAM is used to adjust the weight parameters of each central model to adapt the global model to the unseen domain, as shown in Fig. 2(b). First, the data from the unseen domain D_u is subjected to feature extraction using the same feature extractor used during MLA training and mapped to a feature space. Next, these features are used as inputs to the DAM for domain similarity prediction. The output of the module is the similarity between data features from the unseen domain and those

Table 1

The input, output and function of each layer of DAM. C represents the number of image features, n represents the number of images, and k represents the number of clients.

Layer	Conv 1×1	Fully connection	Full connection	Softmax
Input	$1 \times 1 \times c$	$1 \times 1 \times (n/k)$	$1 \times 1 \times ((n/k) / 16)$	$1 \times 1 \times (k - 1)$
Output	$1 \times 1 \times (n/k)$	$1 \times 1 \times ((n/k) / 16)$	$1 \times 1 \times (k - 1)$	$1 \times 1 \times (k - 1)$
Function	Feature squeezing	Feature extraction	Feature matching	Output probability

from each seen domain. Subsequently, this output is used as aggregation weights to weigh each client's local model, to obtain a new global model. Finally, this new global model is used to implement the application of MLA for inference on unseen domain data. The formula for the new global model can be expressed as

$$G(m) = \sum_{i=0}^k D_i^w l_i(m) \quad (4)$$

where D_i^w represents the predicted probability of belonging to the D_i domain.

3.3. Batch-instance style normalization

3.3.1. Background

BN plays a crucial role in CNN by working on the network in batches, preventing gradient vanishing, accelerating training, and addressing the internal covariate shift problem [41]. IN is a deep learning technique used to normalize each instance in feature channels, which can realize normalization of fixed information of channels in the network [11]. Fine-grained information, such as styles and colors that are fixed attributes in one image, are mainly processed by the shallow layers of CNN. Therefore, using IN in the shallow layers of the network can achieve image style normalization. The advantage of IN is that it can unify styles among several domains, even without other domain data distributions. Therefore, IN can be well applied in FL.

3.3.2. Batch-Instance normalization block

BN alone cannot achieve style normalization for image segmentation and may even lead to feature bias towards local domains, which is detrimental to model generalization. Similarly, IN cannot fully leverage intra-domain features. Both techniques are combined to take full advantage of their individual strengths, reducing intra-domain feature shift and achieving inter-domain style normalization. As shown in Fig. 2 (c), the premise of achieving style normalization is that the network performs well in their respective domains. Therefore, in the BIN block, a 3×3 convolution is first performed, followed by BN to ensure that the performance of the network within the domain is not diminished. Next, the feature channels style normalization is implemented by IN. Then, a ReLU activation function is performed. And the whole process is repeated twice in each BIN block, followed by a max-pooling operation. The details of each layer are shown in Table 2. The formula for the BIN block can be expressed as

$$BIN_{sub} = Act(IN(BN(Conv(f; \theta_c); \theta_{BN}))) \quad (5)$$

$$BIN = MP(BIN_{sub_2}(BIN_{sub_1}(f; \theta_{sub_1}); \theta_{sub_2})) \quad (6)$$

where MP , Act , IN , BN and $Conv$ respectively represent the max-pooling layer, ReLU activation function, IN layer, BN layer, convolutional layer. f , θ_c , θ_{BN} , θ_{sub} respectively represent the block input, parameters of convolutional layer, parameters of BN layer and parameters of BIN_{sub} block.

3.3.3. Networks

Previous research [25] has suggested that style information for images is primarily located in the top-3 layers of networks. In the

Table 2

The input, output and function of each layer of BIN. B represents the batch size. H, w, and c represent the image height, width, and depth respectively.

Layer	Top-1	Top-2	Top-3
Input	$b \times h \times w \times c$	$b \times (h/2) \times (w/2) \times 8$	$b \times (h/4) \times (w/4) \times 8$
Output	$b \times (h/2) \times (w/2) \times 8$	$b \times (h/2) \times (w/2) \times 16$	$b \times (h/4) \times (w/4) \times 32$
Function	Learning intra-domain features, style normalization		

experimental section, it is demonstrated by extensive contrast experiments. Therefore, in this paper, the top-3 layers of the network are defined as the shallow layers. The efficient U-Net is selected as the backbone of segmentation network, with BIN block replacing convolutional blocks in the top-3 layers of the U-Net encoder, while retaining the original convolutional blocks in the fourth and fifth layers to construct the BIN-Net.

3.4. Training and test

3.4.1. Training

For the segmentation networks of each client, the Dice loss is used for training, expressed as

$$L_{seg} = 1 - \frac{2|P \cap G|}{|P| + |G|} \quad (7)$$

where P represents the output of the model and G represents the ground truth under one-hot encoding.

The DAM in MLA is a neural network consisting of multiple perceptrons that are used for classifying image features to identify the domain to which the image belongs. This classification task is trained using the cross-entropy loss function, expressed as

$$L_{cls} = - \sum_{i=0}^{k-1} d_i \log(d_i^p) \quad (8)$$

where d_i represents the label of the domain to which an image belongs, and d_i^p represents the probability predicted by the network that the image belongs to that domain.

3.4.2. Test

During the inference phase, features are firstly extracted from the images of unseen domain and mapped to feature space. Then, the features are inputted into the DAM of MLA to assess the similarity between the image and seen domains. The outputs are used as weight parameters to aggregate the local models and then a new global model adapted to the unseen domain is obtained. Finally, the images are taken into the new global model as the input, where features are extracted by three layers of BIN modules and two convolutional layers and then five upsampling layers to obtain the segmentation prediction of the target organ in the images.

4. Experiments

The proposed MLA-BIN, a combination of MLA and BIN-Net, was extensively evaluated against baseline and state-of-the-art DG models of FL on three multicenter tasks. The tasks include segmentation of the prostate using MRI, segmentation of optic disc and cup (OC/OD) using color fundus images, and lung lesion segmentation using CT scans. The multicenter segmentation tasks cover different modalities of medical images and represent the DG challenges that exist in FL.

4.1. Dataset

Three datasets were employed in this study. For prostate segmentation, a public dataset consisting of 116 T2-weighted MRI cases from six different centers [42], with case ratios of 30:30:19:13:12:12 for each center was used. To ensure consistency in evaluation, only MRI slices with prostate regions were retained for network training and testing. For OC/OD segmentation, a public dataset of 1070 retinal fundus images from four different centers was used [43–46], with case ratios of 101:159:400:400 for each center. Each retina image was cropped to a patch size of 384×384 as input. For lung lesion segmentation, a public dataset consisting of 120 COVID-19 positive cases from four different centers [47,48], with case ratios of 28:19:58:15 for each center was

used. Each CT image was cropped to a patch size of 384×384 as input. These datasets include images and pixel-level target area labels. All data were z-score normalized. In studies [8,15], it has been demonstrated that there exist significant distribution differences between the three datasets, making them suitable for research on DG problems.

4.2. Implementation details

During the FL training process, the same hyperparameters were used for model training across all clients. The batch size for the local models was set to 8, and Adam optimizer was used under default settings. The learning rate was set to 0.01, with a decay step of 100 and a decay rate of 0.8. Each local model was trained for 1000 epochs, and the global model was updated after each epoch to ensure the centralization of model parameters. The DAM of MLA used the same settings as FL, except that the learning rate was set to 0.001 and the epoch was set to 100. Our method was implemented using the PyTorch and trained on a workstation with two NVIDIA V100 GPUs.

4.3. Comparative experiments and analysis

The proposed method was compared with four state-of-the-art DG methods, including (1) BigAug method based on extensive data augmentation [10], (2) ELCFS method based on the principle of amplitude spectrum and meta-learning [7], (3) FedADG method based on federated adversarial learning [6], and (4) DCAC method based on domain and content adaptive convolution [23]. Except for the DCAC method, all other methods used the U-Net as the backbone for model training in the FL framework. The baseline model was set up using the FedAvg [49] algorithm, and it was compared with other methods.

For each segmentation task, datasets from $k + 1$ centers were used, among which each dataset was used in turn as the unseen domain for performance evaluation. The remaining k datasets were used as seen domains for model training. By looping k rounds, each dataset could be evaluated once as the unseen domain. The segmentation results were evaluated using the Dice Similarity Coefficient (DSC) and Average Surface Distance (ASD). DSC (%) and HD (pixels) were used to assess the accuracy of the content and boundary segmentation results, respectively.

4.3.1. Comparative results in prostate segmentation

Table 3 presents the quantitative results of DG for the prostate segmentation task under the FL paradigm. The results indicate that while the BigAug method outperforms the baseline model, it still falls short compared to other DG methods. This suggests that data augmentation methods lack directionality and may not accurately generalize data to unseen domains. The ELCFS method performs better than the FedADG method but slightly lags behind the DCAC method, indicating that although frequency space interpolation extends the distribution of the local dataset, it still lacks adaptability in unseen domains. In contrast, our proposed method achieves an average Dice improvement of 3.08 % over the baseline model. While our method performs slightly lower than other methods on a small number of individual domains, it offers advantages in privacy protection compared to the DCAC method, which

requires data sharing. Importantly, compared to four state-of-the-art DG methods, our proposed method demonstrates superior performance in average Dice and average ASD. This suggests that our method effectively adapts unseen domain images to seen domain models, while mitigating style differences between different domains and improving the model's generalization ability in unseen domains.

4.3.2. Comparative results in OC/OD and COVID-19 lesion segmentation

Tables 4 and 5 present the quantitative results of DG for OC/OD and COVID-19 lesion segmentation under the FL paradigm, respectively.

For the OC/OD segmentation task, the use of undirected data augmentation methods such as BigAug did not yield significant improvements. In contrast, the FedADG method performed slightly better than the BigAug and ELCFS methods, showcasing the advantages of utilizing GAN for feature distribution alignment. The ELCFS method demonstrated strong performance on certain central data, possibly due to the similarity between the generated data and the data in the unseen domain, enabling the model to better adapt to the unseen domain. Compared to these methods, our proposed method exhibits superior adaptability, even in cases of limited data quality, and demonstrates strong generalization to unseen domains. Moreover, all methods display lower model performance on an unseen domain B compared to other unseen domains, attributable to differences in data quality between domain B and the seen domains. However, in such cases, MLA-BIN still outperforms other existing methods.

For the COVID-19 lesion segmentation task, the domain style differences in CT images are relatively small, which reduces factors that weaken the model's generalization capability. This is particularly evident in DG methods that emphasize improving image style differences, such as BigAug and ELCFS. In contrast, such as FedADG that aligns feature distribution, DCAC that adapts to content features and our method that performs domain recognition on the deep layer features of the image, these methods can address potential factors contributing to DG, and therefore outperform methods that only address style differences. Our method, on the other hand, takes into account both style differences and feature differences, resulting in superior segmentation performance.

4.3.3. Visualization results

In Fig. 3, a qualitative analysis was conducted to compare the DG performance of our proposed method and four state-of-the-art methods on different segmentation tasks. The results of each segmentation task, whether for content segmentation or boundary segmentation, consistently show that our method produces more accurate segmentation results. This provides further evidence for the effectiveness of our method in the DG of FL.

Regarding content, variations in the image features extracted by different local models can be attributed to factors such as parameters and training data. These differences can result in inconsistencies in the representation of target objects or regions across different images or domains, which can in turn affect the quality of segmentation. The MLA method addresses this issue by leveraging deep feature recognition and similarity with seen domains to dynamically assign weights. By allocating appropriate weights to different domains based on this

Table 3

Performance of our method, baseline model and four DG methods in prostate segmentation.

Unseen site	A		B		C		D		E		F		Average	
	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓
BigAug [12]	89.73	1.77	85.84	2.14	84.29	2.27	88.29	1.33	83.15	3.23	89.27	1.12	86.76	1.98
ELCFS [15]	90.62	1.16	86.49	1.53	85.43	1.59	89.73	1.08	83.28	3.18	89.18	1.04	87.46	1.60
FedADG [16]	89.92	1.96	86.58	1.78	84.85	2.01	89.38	1.15	83.96	2.17	89.64	1.02	87.39	1.68
DCAC [8]	90.56	0.95	89.54	1.16	84.76	2.16	89.41	1.12	83.54	2.49	89.36	1.25	87.91	1.53
FedAvg [46]	88.36	2.12	84.80	2.31	83.48	2.58	86.42	2.61	79.85	4.49	88.22	1.87	85.19	2.66
Ours (MLA-BIN)	91.98	0.85	87.31	1.27	85.03	1.71	90.42	1.06	84.19	2.29	90.67	0.93	88.27	1.35

Table 4

Performance (OC, OD) of our method, baseline model and four DG methods in OC/OD segmentation.

Unseen site	A		B		C		D		Average	
	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓
BigAug [12]	(81.76, 93.68)	(19.61, 9.16)	(72.49, 86.73)	(25.34, 18.71)	(84.15, 93.42)	(11.24, 9.08)	(84.23, 95.42)	(8.74, 5.66)	86.49	13.44
ELCFS [15]	(84.75, 94.26)	(17.32, 8.77)	(72.35, 86.40)	(25.57, 18.56)	(84.81, 94.26)	(10.73, 8.21)	(85.07, 96.42)	(8.12, 4.97)	87.29	12.78
FedADG [16]	(82.96, 94.47)	(18.85, 8.65)	(73.49, 86.95)	(24.51, 17.93)	(84.29, 95.42)	(10.86, 7.87)	(85.28, 95.76)	(8.03, 5.59)	87.33	12.79
DCAC [8]	(83.17, 94.72)	(18.73, 8.42)	(74.08, 87.66)	(23.27, 17.52)	(84.75, 95.75)	(10.44, 7.62)	(86.89, 96.71)	(7.51, 4.88)	87.97	12.30
FedAvg [46]	(81.06, 92.90)	(20.42, 9.85)	(71.64, 84.26)	(26.19, 22.45)	(83.06, 93.12)	(11.75, 9.35)	(83.15, 95.57)	(9.23, 5.70)	85.60	14.37
Ours (MLA–BIN)	(83.49, 95.15)	(17.81, 8.01)	(74.16, 88.80)	(23.12, 16.38)	(85.57, 96.33)	(10.12, 7.43)	(86.29, 96.23)	(7.87, 5.13)	88.25	11.98

Table 5

Performance of our method, baseline model and four DG methods in COVID-19 lesion segmentation.

Unseen site	A		B		C		D		Average	
	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓
BigAug [12]	63.18	14.67	60.49	15.37	66.34	13.70	63.22	15.42	63.31	14.79
ELCFS [15]	64.19	14.35	61.85	16.09	65.48	12.35	63.86	14.56	63.85	14.34
FedADG [16]	64.24	12.06	61.82	17.35	66.72	13.19	64.39	12.72	64.29	13.83
DCAC [8]	64.73	13.42	62.03	14.64	66.75	12.89	65.10	9.24	64.65	12.54
FedAvg [46]	61.68	15.26	57.15	19.41	63.98	14.83	61.94	15.85	61.19	16.34
Ours (MLA–BIN)	64.56	13.85	62.17	14.03	67.54	11.37	65.48	10.39	64.94	12.41

information, our method allows the model to focus more on relevant features while attenuating the influence of dissimilar features. Consequently, the MLA enhances the model's ability to perform high-quality content segmentation.

As for boundary, style differences can lead to variations in the delineation of object boundary in an image, making it challenging for the network to accurately define object boundary. The BIN module in our method effectively harmonizes style variations in the input images. This enables the network to establish a consistent and standardized criterion for identifying object boundaries, leading to improved segmentation performance. Consequently, the BIN module enhances the model's ability to accurately capture and delineate object boundaries, contributing to enhanced segmentation results. By effectively eliminating style differences, the BIN module unifies the criterion for object boundaries and further improves segmentation performance.

In the seventh and eighth rows of Fig. 3, the OC/OD segmentation task is performed on the dataset from Site B. Consistent with the quantitative analysis results in Table 4, the overall segmentation performance on this dataset is relatively poor. Specifically, our method fails to achieve satisfactory performance in segmenting the OC. This may be attributed to two reasons: firstly, the dataset from Site B itself has lower quality, leading to ambiguous definition of the OC boundary, resulting in inadequate shape representation by our method. Secondly, the dataset from Site B exhibits significant differences compared to the other three sites, and our method heavily relies on the data features from the seen sites, which leads to suboptimal segmentation results in this case.

5. Discussion

We conducted supplementary experiments using the MRI prostate segmentation task as an example to investigate the following issues of our method: 1) the contribution of MLA and BIN to model performance respectively, 2) the comparison of model performance between BIN and other modules, 3) the optimal number of the network's shallow layers, 4) the impact of the number of clients, 5) cost analysis, 6) the impact of local training epochs on the model, 7) using other backbones, and 8) limitations.

5.1. Ablation analysis

In this study, MLA and BIN modules were designed to enable the model to adapt to unseen domain data. To evaluate the contributions of these two modules, in MRI prostate segmentation the model performance is compared with only MLA, only BIN, and both modules used simultaneously, using FedAvg as the baseline model.

The MLA module tackles the problem of domain feature transfer by dynamically assigning weights based on deep feature similarity with seen domains. By considering the unseen perspective, it allows the model to adaptively utilize relevant features from different domains, thereby improving its ability to generalize to unseen domains. On the other hand, the BIN module addresses style differences by considering the visible perspective, thus improving its performance in style-diverse scenarios. As shown in Fig. 4, both MLA and BIN modules can improve the generalization performance on the unseen domain. It indicates that there are domain feature shifts and style differences in DG, and MLA and BIN modules respectively address these two issues to improve the model's generalization performance.

Simultaneously using MLA and BIN modules can compensate for the shortcomings of using a single component and further improve the model's generalization performance. The MLA leverages the concept of multi-model adaptation to learn the differences in domain-specific features across different domains. By doing so, it effectively captures the information specific to each domain while minimizing the adverse effects of domain shift. On the other hand, the BIN addresses domain shift by normalizing feature statistics across domains, thereby aiding in reducing the distribution gap between the source and target domains.

5.2. BIN or other style normalization techniques

The BIN module proposed in this paper has a similar structure to the IBN module proposed by [24], BIN1 module proposed by [50] and D-BIN module by [51]. To facilitate a thorough evaluation of these modules, we employed them individually to train the model in conjunction with MLA and conducted a quantitative analysis in MRI prostate segmentation. In Fig. 5, the quantitative analysis results of these methods and baseline in the prostate segmentation task more directly reflect the differences in model generalization. Obviously, MLA–BIN was superior to other methods with similar structures. This is because the IBN module's



Fig. 3. Visualization of the results predicted by ours (MLA-BIN) and four state-of-the-art methods on the three segmentation tasks, together with ground truth and baseline.

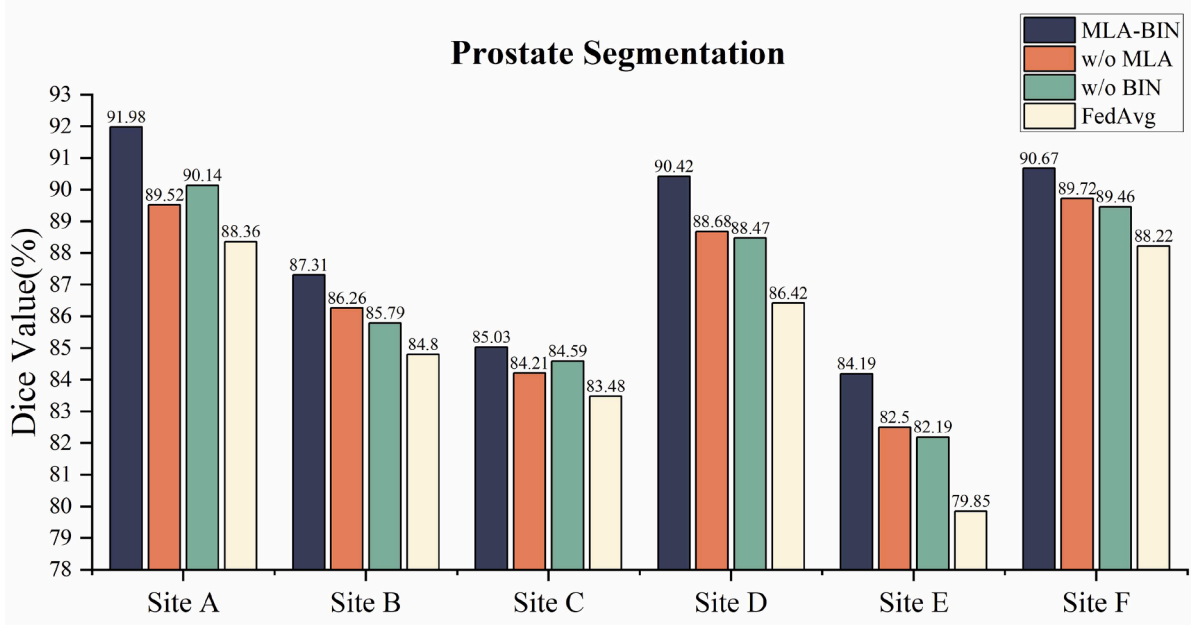


Fig. 4. Ablation results to analyze the effect of the two components (i.e., MLA and BIN) in our method, together with the baseline method (FedAvg).

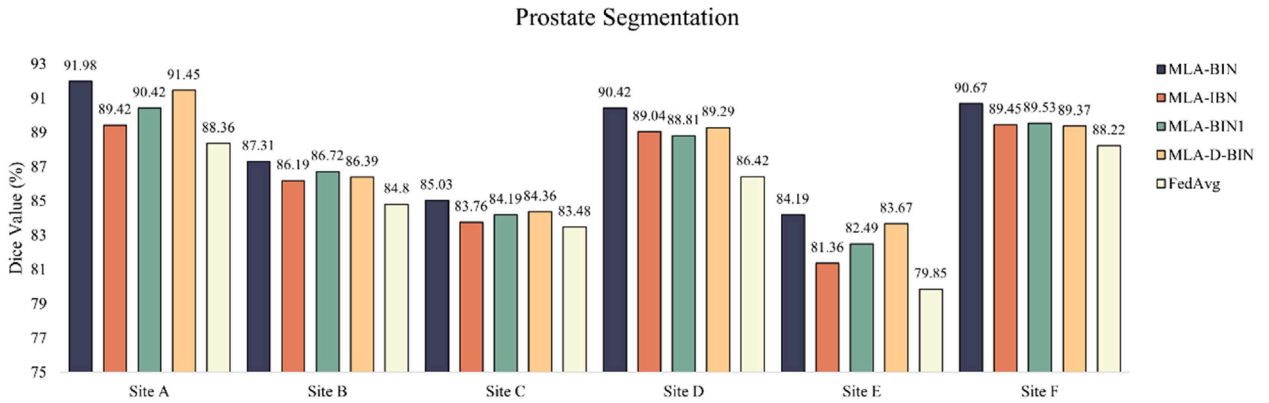


Fig. 5. Performance comparison of our MLA-BIN, MLA-IBN, MLA-BIN1, MLA-D-BIN and the baseline method in prostate segmentation.

strategy of concatenating IN and BN layers to solve style normalization can lead to incomplete style normalization, with the style differences that BN layers retain being transmitted to the next convolutional layer as features. A common characteristic of both BIN1 and D-BIN modules is their ability to adjust the degree of style and content preservation through parameter tuning. However, this approach alone does not eliminate style differences. Style preservation in style normalization can potentially misguide the network. The BIN module avoids this situation by adopting a strategy of linking BN and IN layers. Using IN after BN not only ensures the effective learning of intra-domain features but also fully achieves style normalization, thus avoiding inter-domain style differences. Accordingly, our BIN module not only possesses a more concise structure but also achieves superior performance.

5.3. Optimization of the number of the network's shallow layers

In [24], the shallow layers of the network were defined as the top-3 layers using feature divergence analysis. In our experiments, the number of shallow layers is corroborated from a different perspective by investigating the generalization performance of models with a BIN module embedded into the network from the top-1 layer to the top-5 layers, to find the optimal number of the shallow layers of the network. As shown

in Fig. 6, in MRI prostate segmentation, the best generalization performance is achieved when the BIN module is applied to the top-3 layers, while using the BIN module excessively or insufficiently results in weaker generalization performance than that of embedding the module in the first three layers. A possible reason is that using the BIN module only in the top-1 layer or top2 layers leads to incomplete normalization of image style differences across domains. Moreover, adding the BIN module on the basis of top-3 layers results in the filtration of deep features in the network, which is not conducive to complementary learning with MLA. Therefore, defining the shallow layers of the network as the top-3 layers and applying the BIN module for style normalization in these layers is optimal.

5.4. Impact of the number of clients

In addition, 1 to K-1 domains are randomly selected as training clients in seen domains, and the impact of different numbers of clients on the generalization of FL is analyzed. Fig. 7 shows the generalization performance of Site A and Site B for the prostate segmentation task, using the MLA-BIN and FedAvg methods, with varying numbers of clients. With the increasing number of clients, the performance of MLA-BIN and FedAvg gradually improves. The improvement can be

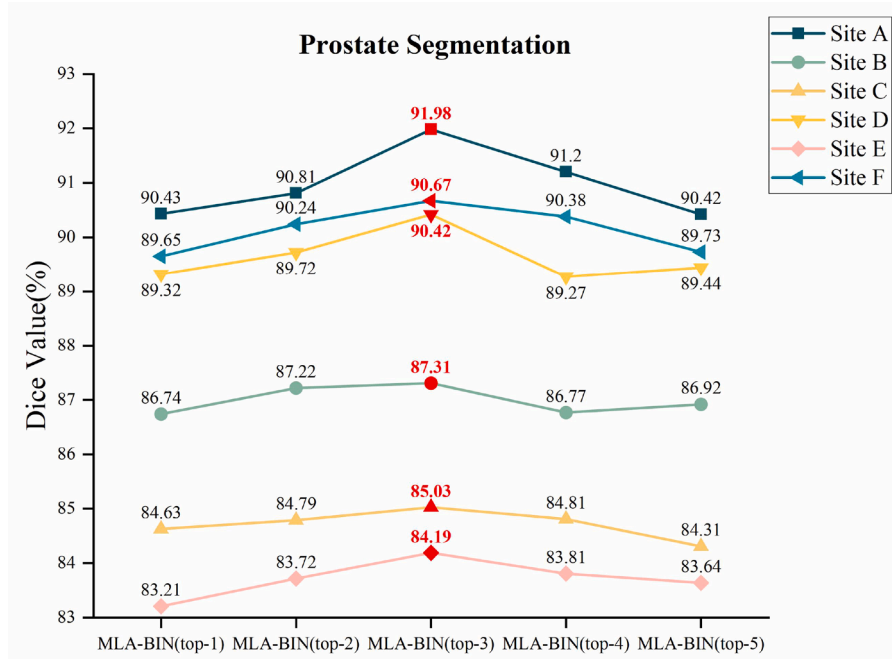


Fig. 6. Performance of MLA with BIN in the top-1 layer to top-5 layers in the prostate segmentation task.

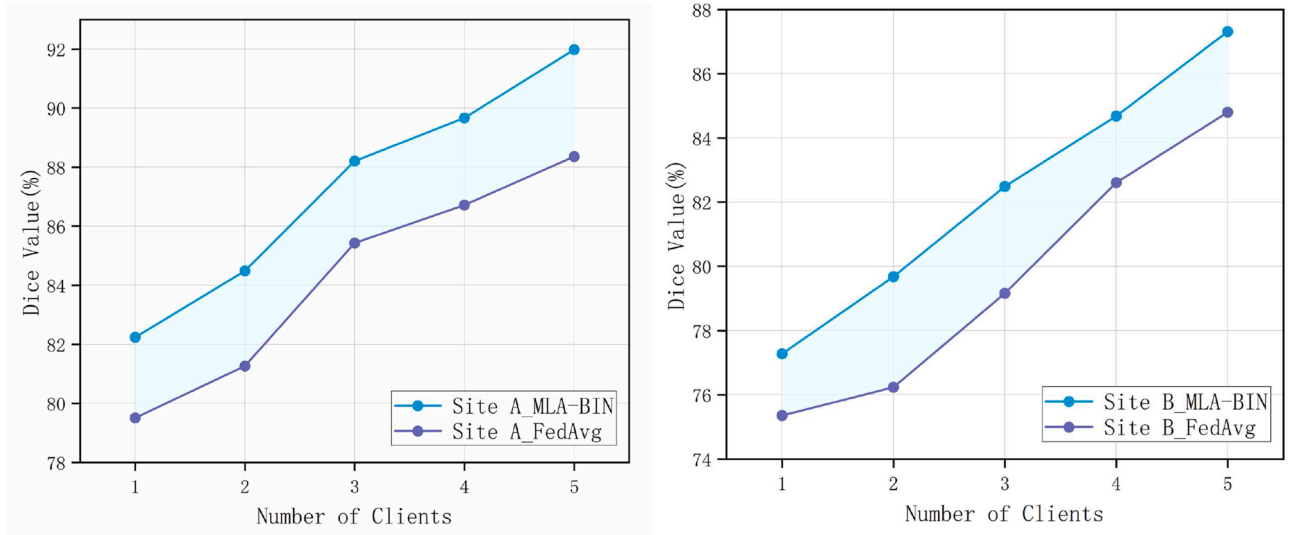


Fig. 7. Generalization performance of models trained with different numbers of client sites with site A and site B serving as unseen domains, using MLA-BIN and FedAvg.

attributed to the increased number of clients, which enables a more comprehensive and accurate representation of known domains in relation to the unseen domain. Furthermore, regardless of the number of clients used, MLA-BIN consistently outperforms FedAvg in terms of generalization performance, which further shows the effectiveness of our MLA-BIN method in the DG of FL.

5.5. Cost analysis

As shown in Table 6, our proposed method incorporates the BIN and MLA modules on top of the baseline model, and it is noteworthy that both modules are lightweight. The BIN block is employed only in the shallow layers of the network, resulting in minimal increases in parameter

Table 6

Number of parameters, model size, GFLOPs and training cost of different models in prostate segmentation.

Method	Parameters ($\times 10^6$)	Model size (MB)	GFLOPs	Training dataset size (GB)	Training time (Hours)	Testing time (s/epoch)
ELCFS [15]	35.44	133.69	74.36	1.18	6.25	30.52
FedADG [16]	55.97	202.18	50.18	1.18	6.75	30.31
FedAvg	34.52	131.74	74.08	1.18	5.32	30.24
+BIN	34.53 (+0.01)	131.75 (+0.01)	74.30 (+2.2)	1.18(+0)	5.32(+0)	30.35(+0.11)
+MLA	35.08(+0.55)	133.88(+2.13)	74.56 (+2.6)	1.20(+0.02)	5.42(+0.1)	30.41(+0.06)

number, GFLOPs, testing time and model size, while not affecting the dataset size or training time. MLA is a lightweight module composed of multi-layer perceptron, which can effectively execute even in CPU environments. Therefore, there is no need to consider the computational burden on the server when training MLA. As a pre-trained model, the MLA is directly applied during testing, requiring only inference parameters and resulting in minimal increase in testing time. This reduces the need to consider time costs in clinical deployment.

Furthermore, the increase in training dataset size mainly stems from the need for MLA to share domain features and corresponding labels with the server, which are stored in the domain feature bank. Fortunately, these features and labels are $n \times 1025$ -dimensional vectors, where n represents the number of client images. Compared to the size of the images, this is relatively small and only needs to be transmitted once during initialization. Hence, it does not impose additional storage and communication costs that are burdensome for FL.

Additionally, on one hand, the domain feature bank stores features rather than original images, and on the other hand, because its usage is limited to the training phase and it requires short training time, the domain feature bank can be viewed as a transit station. The stored data can be destroyed once training is completed. In practical deployment, pre-trained MLA can be directly used, eliminating concerns about the risk of data leakage.

Compared to other models, our method demonstrates shorter training time compared to ELCFS, while exhibiting fewer parameters, model size, GFLOPs, and training time compared to FedADG. This indicates that the approach of GANs incurs additional computational overhead. Thus, MLA-BIN achieves superior model performance with smaller space and computational complexity.

5.6. Impact of local training epochs on the model

In this experiment, we investigated the impact of varying numbers of local training epochs on the convergence and generalization capabilities of the model. We conducted a series of experiments to train the model with different local training epochs (1, 2, 5, 10) for aggregation. Fig. 8 illustrates the convergence and generalization performance of the MLA-BIN method for the prostate segmentation task of Site A and Site B, considering different aggregation rounds. The results of our experiments revealed interesting insights into the effects of local training epochs and aggregation rounds on model convergence and generalization. We observed that increasing the number of local training epochs led to improved convergence, as the model had more opportunities to learn domain-specific features. However, it should be noted that there was a decrease in model performance with the decrease of aggregation frequency, indicating the presence of overfitting and excessively long local training epochs. Regarding aggregation rounds, we found that aggregating after each local training round contributed to better cross-domain generalization. This suggests that increasing the inter-domain communication frequency helps mitigate the impact of noisy updates from local

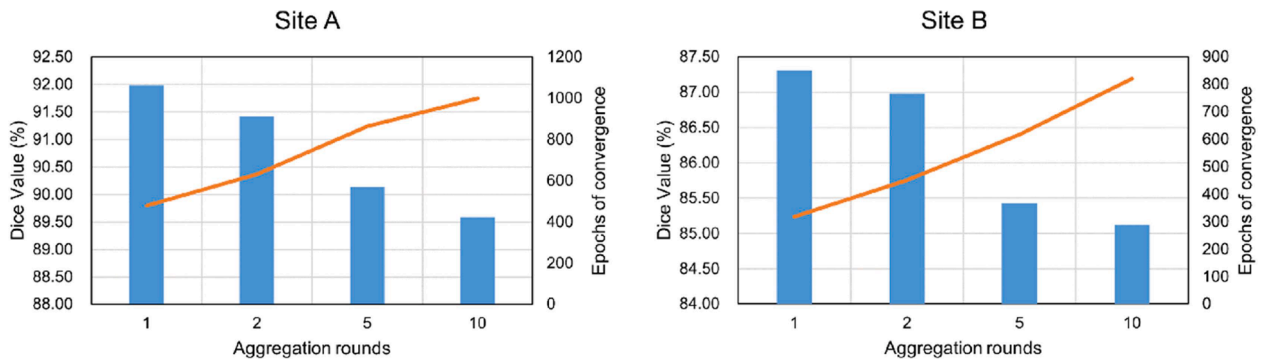


Fig. 8. Convergence and generalization performance of MLA-BIN methods with different aggregation rounds for prostate segmentation tasks of site A and site B.

training and enhances the model's ability to generalize to unseen data. Overall, our findings emphasize the importance of balancing the number of local training epochs and aggregation rounds to achieve the optimal model convergence and generalization. These insights can guide the fine-tuning of hyperparameters in domain adaptation models, leading to improved performance of DG.

5.7. Using other backbones

The proposed MLA-BIN method, as a plug-and-play module, can be integrated into other segmentation networks to enhance the model's generalization performance. To validate this claim, we employed U-Net++ [52] as the segmentation backbone and integrated MLA and BIN modules within the backbone for the prostate segmentation task. Fig. 9 illustrates the generalization performance of the U-Net++ method with and without the MLA-BIN on six domains. The results indicate a consistent enhancement in the generalization ability of the U-Net++ model equipped with the MLA-BIN module, across diverse sites. This provides compelling evidence for the compatibility and efficacy of our proposed method.

5.8. Limitations

First, like the limitation of FL, the aggregation of FL model has uncertainty. In other words, the effect of new aggregation in the next round is not necessarily better than the previous round. However, we believe that making the aggregation of FL more feedback-capable is beneficial for improving the performance of the model. Second, as evident from the results of unseen domain B in the OC/OD segmentation task, acquiring more client data is essential to improve the model's fitting to unseen domains, which has been confirmed in the discussions. However, it is common for them to be a scarcity of source domains and a multitude of

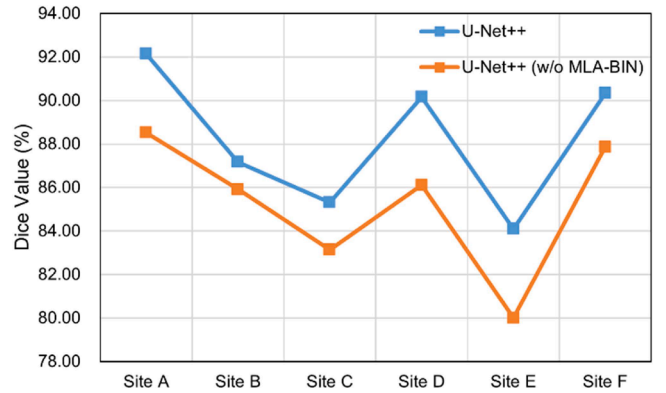


Fig. 9. Performance of U-Net++ with and without MLA-BIN on prostate segmentation task.

unseen domains, posing a shared challenge for domain generalization—how to enhance model performance in such circumstances. Third, due to the requirement of coordination between intra-domain and inter-domain learning in our proposed method, the model is susceptible to the influence of hyperparameters such as batch size. Consequently, it necessitates careful tuning of the training model. In our future work, we will incorporate prior knowledge to mitigate significant disparities between predicted results and the ground truth. Furthermore, we will introduce unsupervised learning to reduce reliance on seen domains, thereby addressing the limitations of the proposed method.

6. Conclusion

This paper proposes a novel method to address DG problems in FL by incorporating a MLA module with a BIN module. The MLA represents an unseen domain as a linear combination of seen domain models and adaptively adjusts the weights of seen domain models based on the similarity of deep layer features between unseen and seen domains. The BIN, which combines BN and IN, is applied to the shallow layers of the network to achieve effective learning of intra-domain features and inter-domain style unification. Our results in three multi-center medical image segmentation tasks demonstrate that our proposed method can effectively generalize from seen to unseen domains in the FL paradigm and outperforms state-of-the-art DG methods.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to check for syntactic issues. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRedit authorship contribution statement

Fubao Zhu: Conceptualization, Formal analysis, Funding acquisition, Methodology, Resources, Writing – original draft. **Yanhui Tian:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft. **Chuang Han:** Funding acquisition, Validation, Writing – review & editing. **Yanting Li:** Funding acquisition, Supervision, Validation, Writing – review & editing. **Jiaofen Nan:** Funding acquisition, Supervision, Validation, Writing – review & editing. **Ni Yao:** Investigation, Resources, Supervision, Writing – review & editing. **Weihua Zhou:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this paper have been indicated and annotated in the references.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.inffus.2024.102348.

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